

## 物理域 $r$ (随时间收缩)

$$t_0 \quad R(t_0) \approx 1.8 \text{ cm}$$



$$t_1 \quad R(t_1) \approx 1.4 \text{ cm}$$



$R(t) \downarrow$

$$t_{\text{dry}} \quad R(t_{\text{dry}}) \approx 1.08 \text{ cm}$$



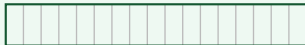
网格随收缩同步加密

坐标变换

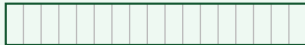
$$\xi = \frac{r}{R(t)}$$



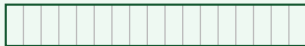
## 计算域 $\xi = r/R(t)$ (固定)



$\xi \in [0, 1]$ , 不随时间变化  
 $\Delta\xi = 1/160$ , 节点数不变



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$$R(t) : 2.000 \rightarrow 1.200 \text{ cm} \quad | \quad N = 160 \quad (\Delta\xi = 1/160) \quad | \quad R(t)$$